

E-Commerce Platform — Product Requirements Document

ShopSphere

Field	Value
Document version	1.0
Status	Draft for stakeholder review
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Target stack	React 18 (Vite) + Node.js 20 (Express) + PostgreSQL + Redis

1. Executive Summary

ShopSphere is a modern, mobile-first e-commerce platform enabling merchants to sell physical products online. The MVP targets a single-tenant storefront with a full customer purchase journey (browse → cart → checkout → order tracking) and an admin console for catalog, inventory, and order management. The system is architected with Domain-Driven Design so that the MVP monolith can be decomposed into services (Catalog, Ordering, Payments, Fulfilment) as scale demands.

2. Goals & Non-Goals

2.1 Goals

- Launch a production-ready storefront supporting 10k daily active users at p95 page latency < 300 ms.
- Checkout conversion rate $\geq 3\%$ within 3 months of launch.
- Cart abandonment recovery via persisted carts and email reminders.
- Admin can create a product and see it live on the storefront in under 2 minutes.
- PCI-DSS scope minimization by delegating card handling to a hosted payment provider (Stripe/Razorpay).

2.2 Non-Goals (MVP)

- Multi-vendor marketplace, multi-currency pricing, and B2B quoting.
- Native mobile apps (the responsive web app is the mobile experience).
- In-house payment processing or card vaulting.
- Warehouse management beyond simple stock counts.

3. Personas

Persona	Description	Primary Jobs-to-be-Done
Shopper (guest)	First-time visitor, mostly mobile	Discover products, compare, buy without creating an account
Shopper (registered)	Returning customer	Reorder quickly, track orders, manage addresses and wishlist
Store Admin	Merchant operations staff	Manage catalog, prices, stock, discounts; process and refund orders
Support Agent	Customer service	Look up orders, trigger refunds/returns, edit shipping details

4. User Stories & Acceptance Criteria (MVP)

ID	Story	Acceptance criteria (summary)
US-01	As a shopper, I can browse products by category with filters and sorting.	Filter by price range, brand, rating; sort by price, newest, popularity; paginated (24/page); SSR-friendly URLs.

ID	Story	Acceptance criteria (summary)
US-02	As a shopper, I can search products with typo tolerance.	Search-as-you-type < 150 ms; fuzzy matching; empty-state suggestions.
US-03	As a shopper, I can view a product with images, variants, stock and reviews.	Variant selection updates price/stock/SKU; out-of-stock variants disabled; image zoom and gallery.
US-04	As a shopper, I can add items to a cart that persists.	Guest cart in localStorage merged into account cart on login; line-item stock revalidated at checkout.
US-05	As a shopper, I can check out as guest or registered user.	Address form with validation; shipping method selection; payment via hosted provider; idempotent order creation.
US-06	As a shopper, I receive order confirmation and can track status.	Email within 1 min; statuses: Placed → Confirmed → Shipped → Delivered / Cancelled.
US-07	As an admin, I can CRUD products, variants, and inventory.	Draft/published states; bulk stock update via CSV; audit log of changes.
US-08	As an admin, I can view and manage orders.	Search by order #, email; change status; full/partial refunds via payment provider API.
US-09	As an admin, I can create discount codes.	Percentage/fixed amount; validity window; usage limits; min cart value.
US-10	As a shopper, I can review purchased products.	Only verified purchasers; 1–5 stars + text; admin moderation queue.

5. Functional Scope by Release

Release	Scope
R1 — MVP (Weeks 1–10)	Catalog, search, cart, guest+registered checkout, Stripe/Razorpay payment, order emails, admin catalog & order management, discounts.
R2 (Weeks 11–16)	Reviews & ratings, wishlist, cart-abandonment emails, returns/RMA flow, basic analytics dashboard.
R3 (Weeks 17+)	Recommendations, multi-warehouse inventory, CMS-driven landing pages, loyalty points.

6. Non-Functional Requirements

Category	Requirement
Performance	p95 API latency < 200 ms; LCP < 2.5 s on 4G; catalog reads cached in Redis with 60 s TTL.
Scalability	Stateless Node processes behind a load balancer; horizontal scale to 50 req/s sustained; DB read replicas for catalog.
Availability	99.9% monthly uptime; graceful degradation: checkout stays up even if search/reviews are down.
Security	OWASP ASVS L2; JWT access (15 min) + rotating refresh tokens (httpOnly); rate limiting; RBAC (shopper/admin/support); no raw card data touches our servers.
Consistency	Orders and payments are strongly consistent (ACID, Postgres transactions + outbox pattern); catalog/search eventually consistent.
Compliance	GDPR/DPDP data export & deletion; PCI SAQ-A via hosted payment fields.
Observability	Structured JSON logs, OpenTelemetry traces, RED metrics per endpoint, alerting on checkout error rate > 1%.
Accessibility	WCAG 2.1 AA on all storefront pages.

7. System Architecture (Summary)

The MVP ships as a modular monolith: a single Node.js/Express deployable internally partitioned into DDD bounded contexts (Identity, Catalog, Ordering, Payments, Fulfilment, Promotions). React SPA served via CDN; API is versioned REST (/api/v1). PostgreSQL is the system of record; Redis handles sessions, cart cache, and rate limiting; a transactional outbox with a worker publishes domain events (OrderPlaced, PaymentCaptured, StockDepleted) to enable async email, inventory, and analytics flows and a clean path to future service extraction. Full details are in the companion DDD document.

8. Key Metrics

- North star: weekly completed orders.
- Conversion funnel: view → add-to-cart → checkout-start → payment-success.
- Cart abandonment rate; recovery-email conversion.

- p95 latency and checkout error rate (SLO-backed).

9. Risks & Mitigations

Risk	Mitigation
Overselling under concurrent checkout	Row-level stock reservation with SELECT ... FOR UPDATE + reservation TTL; idempotency keys on order creation.
Payment webhook loss/duplication	Signed webhooks, idempotent handlers keyed by provider event ID, reconciliation job.
Search relevance poor at MVP	Start with Postgres full-text + trigram; abstract behind SearchPort to swap in OpenSearch later.
Scope creep in admin	Admin MVP limited to catalog, orders, discounts; everything else via runbook/SQL initially.

10. Open Questions

- Payment provider: Stripe (global) vs Razorpay (India-first) — decide by pricing and settlement needs.
- Shipping rates: flat-rate MVP vs live carrier rates (Shiprocket/EasyPost) in R2.
- Tax engine: static GST slabs at MVP vs external tax service.